NOISEMAP · TRACEABILITY MATRIX

NoiseMap traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/noisemap		
Source	docs/03b-traceability.md @ ffa26f8		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-11-02	Amish Chadha	Generated by .kit/trace.py
0.2	2026-11-02	Amish Chadha	Generated by .kit/trace.py
0.1	2026-11-02	Amish Chadha	Generated by .kit/trace.py

NoiseMap traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 16 user requirements, 17 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Levels only	NSM-CAL-001, NSM-BLD-001, NSM-STM-001	Design review, firmware audit, cable capture (TRL 3 design review; audit and capture at TRL 4)	Met by design; the guarantee rests on the firmware, since the cable could carry a speech codec (NSM-DDR-001 D2)
UR-002	PR-002	Reported metrics	NSM-CAL-001, NSM-BLD-001, NSM-STM-001	Design review (TRL 3)	Met by design at SF7 to SF9; sub-intervals longer at SF10 to SF12
UR-003	PR-003	Accuracy	NSM-CAL-001, NSM-STM-001, NSM-OPT-001	Analysis (NSM-CAL-001 C3, TRL 3); side-by-side field test (TRL 4)	Not met on paper: expanded uncertainty 2.9 dB (3.1 dB with the reference meter) on assumed terms against 2 dB; facade bias up to 2.4 dB (2.7 dB in a canyon) before a site correction; relaxing the target is Amish's decision
UR-004	PR-004	Measuring range	NSM-CAL-001, NSM-BLD-001, NSM-STM-001, NSM-OPT-001	Analysis (NSM-CAL-001 A1, TRL 3); bench test (TRL 4)	At risk: ICS-43434 34.9 dBA to 113 dB SPL, which is 103 to 108 dBA for traffic; IM72D128 (fitted by default) 27.9 dBA to 119 dB SPL, 109 to 114 dBA
UR-005	PR-005	Frequency weighting	NSM-CAL-001, NSM-OPT-001	Analysis (NSM-CAL-001 B, TRL 3); bench test (TRL 4)	At risk: roll-off equalized to 0.8 dB at 63 Hz; membrane, windscreen and head diffraction unknown
UR-006	PR-006	Time stamps	NSM-CAL-001	Design review (LoRaWAN network time, TRL 3)	Met by design
UR-007	PR-007	Energy	NSM-CAL-001, NSM-STM-001	Analysis (NSM-CAL-001 D and E, TRL 3); bench test (TRL 4)	Met on paper (5.81 Wh/day for 0.29 Wh/day; 53 days)
UR-010	PR-008	Ingress and weather	NSM-CAL-001, NSM-BLD-001	Spray test and thermal chamber (TRL 4)	To verify at TRL 4
UR-012	PR-009	Calibration check	NSM-CAL-001, NSM-BLD-001, NSM-STM-001	Timed trial (TRL 4)	To verify at TRL 4
UR-008, UR-009	PR-010	Installation, mass and wind	NSM-CAL-001, NSM-BLD-001, NSM-STM-001	Analysis (NSM-CAL-001 H, TRL 3); timed trial and weighing (TRL 4)	Met on paper on mass (3.45 kg, margin 0.05 kg), wind and fit; install time to verify at TRL 4
UR-011	PR-011		NSM-CAL-001		

User requirement	Product requirement	Title	Evidence	Verification	Status
		Contaminated data		Field comparison with a weather station (TRL 4)	To verify at TRL 4: method chosen, threshold needs field data
UR-013	PR-012	Radio use	NSM-CAL-001, NSM-STM-001	Analysis (NSM-CAL-001 F, TRL 3)	Met on paper: 23.7 s/day at SF9; 29.97 s/day at SF12, with almost no margin for join, time requests and retries
UR-014	PR-013	Parts cost	NSM-CAL-001	Priced BOM (NSM-CAL-001 I, TRL 3)	Met on paper: USD 72.00, USD 28.00 under the target; full node USD 211.00
UR-010	PR-014	Service life	NSM-CAL-001, NSM-STM-001	Design review (TRL 3); ageing test (TRL 4)	To verify at TRL 4
UR-015	PR-015	Openness	NSM-CAL-001	Design review (TRL 3)	Met by design
UR-016	PR-016	Public notice	NSM-BLD-001, NSM-STM-001	Design review (TRL 3); inspection at installation (TRL 4)	To verify at TRL 4: the plate is part of the installation, not of the BOM
UR-009	PR-017	Falling-part retention	NSM-BLD-001, NSM-STM-001	Design review (build plan Step 11, TRL 3); pull test at TRL 4	Met by design (modelled in cad/src/model.py, BOM line 13)

Gaps

None.